

Guide For Engineers To Create INNOVATIVE, MANUFACTURABLE PCBs

Navigating modern PCB design can feel overwhelming. From CAD tools to supplier selection, Part 1 of this series offers actionable insights to simplify your journey.



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The printed circuit board (PCB) industry continues to expand and evolve, with designers now facing heightened challenges post-Covid, including component end-of-life issues and supply chain uncertainties. The demand for flawless, first-attempt designs has never been higher.

This series is designed to guide PCB engineers in creating innovative and manufacturable designs while navigating the complexities of modern product development. For those new to PCB design or with limited experience, understanding the principles of 'Designing for Reality' is critical. This approach emphasises robust methodologies that ensure reliability, manufacturability, and compliance with diverse design requirements.

Success depends on meticulous attention to detail. PCB design is shaped

by unwritten rules, best practices, and varying manufacturer-specific requirements. Visiting manufacturing facilities is a valuable step towards grasping the nuances of board production, which can differ significantly depending on the classification and design specifications.

By partnering with reliable manufacturers and developing a deep understanding of production fundamentals, designers can overcome challenges and consistently achieve functional, cost-effective designs—even as projects grow more complex.

A glimpse ahead

This series will explore essential strategies for PCB designers, beginning with selecting the right CAD tools, qualifying suppliers for seamless production, and mastering component placement. Subsequent parts will delve into advanced